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Linking Theories of Motivation, Game Mechanics, and Public Deliberation to Design an Online System for Participatory Budgeting

John Gastil and Michael Broghammer

Abstract

Existing systems for online civic engagement and public consultation need a better architecture if they are to realize the aspirations of deliberative democracy. To improve the design of such systems, we develop an empirical model of online civic engagement that connects common game mechanics to four key democratic processes and outcomes—inclusion, deliberative engagement, sound and influential public input, and long-term civic impacts. We then link game mechanics and deliberation with theories of motivation to show how these mechanics can leverage people's drives to fulfill basic needs, forge social connections, and gain status. To illustrate our model in more concrete terms, we show how game mechanics could motivate both participants and policymakers in an online participatory budgeting system. We conclude by describing a multi-stage experimental approach to testing this model within an existing system of online participatory budgeting.

Keywords

civic engagement, deliberative democracy, game design, motivation, participatory budgeting

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Society's basic communication infrastructure has been transformed thanks to the widespread adoption of advanced digital technology, the proliferation of information sources, and the expansion of social media (Howard, 2015; Sunstein, 2017). Our digital age is both promising and problematic for public deliberation (Nilsson and Carlsson, 2014). New technology has broadened the public sphere to include more voices often in their own enclaves (Simone, 2010), but the net effect of this "accelerated pluralism" may be a

Department of Communication Arts and Sciences and Political Science, Pennsylvania State University, State College, PA, USA

Corresponding author:
John Gastil, Department of Communication Arts and Sciences, Pennsylvania State University, 234 Sparks Building, University Park, PA 16802, USA.
Email: jgastil@psu.edu

more volatile public discourse (Bimber, 1998). Political campaign outreach has become more expansive and intensive which can inspire public participation (Green and Gerber, 2008), but finely targeted messaging can be manipulative by serving purposes invisible to the voters being contacted (Persily, 2017). Governance has become more visible by posting data, records, and meetings online, but the net benefit of these practices remains uncertain at best (Piotrowski, 2017; Porumbescu, 2017).

Particularly contentious is the decades-long debate over the effectiveness and utility of online versus offline attempts at public deliberation (e.g. Baek et al., 2012; Gastil, 2000; Min, 2007; Stromer-Galley, 2002). Many articles in this Special Issue advance this same debate by assessing the threats and opportunities that digital media pose for public deliberation. When viewed in the broader framework of deliberative systems (Parkinson and Mansbridge, 2012), however, the deliberative potential of online systems depends on how they may fit into a larger institutional context (Ercan et al., 2018; Lyons, 2017). The efficacy of any digital deliberative system depends on how it is *designed*.

Thus, one challenge for scholars of politics is to theorize and envision online systems that could better serve democratic purposes. Facebook, Twitter, YouTube, and myriad other "civic media" (Gordon and Mihailidis, 2016) constitute an ever more important part of public life (Howard, 2015; Wells, 2015), yet research on such interfaces has only begun to explore how such interfaces might be harnessed for more deliberative purposes (Davies and Chandler, 2012; Manosevitch, 2014). Discovering what public discourse is possible in such spaces requires varying their core design features systematically and testing them under conditions with real political stakes.

To address that need, we propose a new model of digital deliberative civic engagement that draws on key elements of game design—a process sometimes called "gamification" (Chou, 2015). In the broadest sense, to "gamify" something means the application of techniques, tools, and lessons from games. The result is not necessarily a full-fledged game but something that resembles or evokes games in certain respects. Gamification of an online discussion system, for example, might involve the addition of points for comments that get "liked" or "up-voted" by others, with a leaderboard showing whose contributions have earned the most.

Not every effort to gamify politics and civic engagement works well, and the problems documented thus far include oversimplification of issues, limiting users' autonomy, and demographic differences in participation rates (Meloni et al., 2018; Spada et al., 2016). Even so, past research has found that game mechanics can motivate lay citizens to participate, deliberate, and engage constructively with public institutions while simultaneously motivating agencies and policymakers to be more responsive to public input (Gordon et al., 2017; Hassan and Hamari, 2019; Lerner, 2014; Mayer, 2009; Thiel et al., 2016; Tolmie et al., 2014). Even so, this literature remains under-theorized (Hassan, 2017; Morschheuser et al., 2017). Specific game elements are not clearly connected to specific outcomes, and studies have not clarified the mediating mechanisms through which these game elements operate (Boyle et al., 2016; Giessen, 2015; Thiel et al., 2016). Thus, we hope to propose a plausible means for improving online deliberative spaces while simultaneously advancing theories of game design and motivation linked to deliberative engagement.

Our argument unfolds in four stages. First, we outline participatory budgeting (PB) as a fruitful research context for developing, elucidating, and testing a model of how digital tools can motivate deliberative public engagement. Second, we explore the conceptual connections among game design, democracy, and online engagement, and we show the relevance of such theoretical links to PB programs. Third, we advance a motivational

model of online engagement that shows how online game mechanics can motivate citizens and public officials to produce more deliberative processes and outcomes. We conclude by arguing for field experiments as the best means of testing our propositions regarding the use of gamified online tools in existing PB systems.

PB as a Research Setting

We focus on PB as an ideal research setting. Gamified designs could be observed and tested in a multitude of promising online systems that likely underperform relative to their deliberative potential (Noveck, 2015; Smith, 2018). PB is an ideal research context for three reasons. First, it involves empowered public engagement, so it involves real stakes that can motivate participants to take action. Second, despite its successes, PB has problems with participation, deliberation, and diffusion, which means it has room for improvement. Third, its practitioners have embraced digital technology, which means that there exist opportunities for future experiments using online tools within PB systems.

Empowered Engagement

PB provides the residents in a given jurisdiction a measure of political power by permitting them to conceive ways to spend funds earmarked for their community and vote on their preferred projects. Those proposals that receive the most votes are subsequently funded, and the same residents and civic organizations who participated in the PB process often have the chance to oversee their subsequent implementation (Gilman, 2016).

Past research has stressed the importance of such *empowered* engagement opportunities in part because a well-distributed exercise of power is at the core of democratic self-government (Dahl, 1989). Deliberative and participatory democratic processes have become so ubiquitous that theorists can now discuss their variations in their empowered designs (Johnson and Gastil, 2015), in contrast to often disempowered public consultations (Johnson, 2009, 2015). In addition to being a democratic virtue, empowered engagement may be necessary to obtain other desired goods, such as boosts in civic attitudes and political mobilization that arise from deliberative processes (Myers et al., 2018).

Strong Performance with Room for Improvement

PB is also a useful research context because it has a strong track record, yet needs improvement. PB has proven an adequate means of empowered governance generating both sensible spending priorities and ancillary civic benefits, such as engaging underrepresented social groups or inspiring ongoing civic engagement (Gilman, 2016). Such successes explain its rapid spread across the globe over the past three decades (Dias, 2018).

Despite its success, PB still warrants design innovation. For example, it has not always improved social equity (Su, 2017), and its emancipatory power may be waning (Baiocchi and Ganuza, 2014). Critics have scrutinized the quality of deliberation that occurs in its decision-making processes, as well as government's capacity to design impartial spaces that maximize deliberative opportunities (Ganuza and Francés, 2012). Others have worried about PB's sustainability (Souza, 2007), as it has shown signs of decline in Brazil (Spada, 2012). Concerns about falling participation levels are one reason that many PB processes have gone online, yet evaluations of online tools used in PB can still record public frustration with the process and outcomes (Barros and Sampaio, 2016).

Embracing Online Engagement

Theoretically, moving PB either partly or fully online may help broaden, deepen, and sustain deliberative civic engagement by reducing the opportunity cost of participation for residents, maintaining their connection to the process over time (through stable cell phone numbers and email addresses), and creating spaces for deliberative civic engagement at every stage of the PB process. Simply "going digital" is no guarantee of success. For instance, one head-to-head comparison of online and offline PB processes in South Korea found the offline process more representative, deliberative, and influential than its online counterpart (Lim and Oh, 2016).

Unrealistic expectations are common when government and civic organizations adopt digital technologies to improve public engagement (e.g. Johnson and Roman, 2015). Those who practice PB have begun to explore in earnest the deliberative potential for new online extensions of a traditionally offline process (Miori and Russo, 2011; Peixoto, 2009), which makes PB a viable research setting for experimentation with online design innovation.

Examples abound of partially gamified online PB processes. For instance, a PB process in Belo Horizonte, Brazil, used an online game with a leveling system to ratchet up the knowledge and skills of younger residents interested in the city budget (Meloni et al., 2018). The *Delivering Denver's Future* program in the United States included concrete choice-points and context-sensitive information feedback for participants to structure their experience of reviewing and adjusting the city's budget (Haller and Faulkner, 2012). The European EMPATIA consortium created a full-fledged game, *Empaville*, which foregrounded the conflicting interests of the characters in the game and asked participants to view the budget from a particular role with its own point of view (Meloni et al., 2018). What such experiments need is a stronger theoretical account of the potential benefits of moving PB online and an experimental method for testing such propositions.

In sum, we have three reasons for using PB as a research setting in which we theorize online designs for public engagement: PB provides participants with an empowered opportunity for exercising their political voice; it has shown its potential as an effective process while also having room for improvement; and many PB designers have already incorporated online elements. Next, we show how game design provides a useful framework for designing online tools for PB.

The Relevance of Game Design Categories to Deliberative Democracy

Online systems for public participation and deliberation involve myriad design choices, such as synchronous versus asynchronous communication and public identities versus anonymity (Berg, 2016; Davies and Chandler, 2012). We do not intend to catalog all these choices but instead focus on the design considerations that fall under the umbrella term of a "gamified" civic engagement (Gordon et al., 2017; Lerner, 2014).

Before discussing gamification, we begin with a caveat. Gamifying a process does not always produce the intended results. Examples of problematic gamification have appeared in multiple countries and contexts, including cases where it was applied to PB (Lerner, 2014; Meloni et al., 2018; Spada et al., 2016). For example, gamified spaces could prove so enticing that they draw too much energy away from other political channels, such as public protest and campaigning (Meloni et al., 2018). That said, there is also substantial

evidence that gamification can prove beneficial in both online and offline spaces. Lab and case studies have shown that different game-inspired features may draw people into such spaces, hold people's attention, and alter their behavior and attitudes—whether to induce purchases, forge social connections, or build bridges across cultural divides (Burak and Parker, 2017; Burke, 2014; Chou, 2015; McGonigal, 2011). Our aim here is not to resolve the pro/con debate on gamification but instead to advance the theoretical understanding of how and why a gamified public space can yield certain outcomes.

Game Design Features

The most elaborate attempt made so far to bring together principles and elements of game design and democratic processes has been Josh Lerner's (2014) *Making democracy fun: how game design can empower citizens and transform politics*. Lerner (2014: 29) defines games as "systems in which players engage in an artificial conflict, defined by rules, that results in measurable outcomes." He posits that applying game design principles and elements to democratic processes can address common problems in modern political institutions: "citizens are engaging in democracy less, trusting it less, and granting it less power. Making democracy more enjoyable can help address each problem" (Lerner, 2014: 204).

Lerner (2014) lays out 26 specific game mechanics, which he organizes into four broad categories: (1) a central conflict or collaborative goal driving play; (2) rules or norms governing play; (3) identifiable outcomes with incremental indicators of progress made toward goals; and (4) an array of features and affordances that make play more enjoyable. Lerner's categorizations serve well as classifications not only for his game mechanics but also for other inventories of game elements developed by others (Burak and Parker, 2017; Chou, 2015; Salen and Zimmerman, 2004; Schell, 2014). We discuss the contents of Lerner's categories in more detail later but must first summarize the intended reasons for deploying such game design tools.

Deliberative Democratic Purposes

To link game elements with deliberation, we cross Lerner's (2014) four categories with four procedural and outcome variables critical to evaluating deliberative processes. In theory, each variable may be leveraged by integrating a specific game design feature into a deliberative process. Among other considerations, deliberative processes are evaluated based on the following: (1) the participants they attract and engage in ongoing discussions; (2) the deliberative quality of such discussions; (3) the quality of the public input generated by such discussions and its influence on political and policy outcomes; and (4) the transformative effect of a deliberative experience on a participant's long-term civic attitude and activity (Gastil et al., 2012, 2017).

The first two variables are input and process variables with inherent value for a deliberative democracy. Effective recruitment and active public participation reflects a basic democratic principle of inclusion (Dahl, 1989). A deliberative process scoring higher on this dimension may attract a larger number and greater diversity of participants with their public participation sustained over time. Process quality measures encompass the very act of democratic deliberation itself. A higher-quality deliberation must exhibit democratic norms that the participants perceive as legitimate, and it must involve rigorous analysis of the problem at hand (Burkhalter et al., 2002; Cohen, 1997; Niemeyer and Dryzek, 2007).

The other two focal variables are outcomes measured in deliberative processes' designs. For many theorists, democratic deliberation must deliver on a democratic promise of self-government by linking a deliberative process to actualized policy output or political outcomes (Barrett et al., 2012; Schattan et al., 2005). At a minimum, this may amount not only to generating actionable, high-quality public input but also to inspiring political action or advocacy on the part of deliberative participants or stakeholders concerned about the issues addressed. In some cases, deliberative participants exercise direct power in the policymaking process, but more often, they can only hope to elicit responses from policymakers.

Apart from such direct effects on politics and policy, deliberative theorists often address the transformative effect experiencing democratic deliberation can have on those participants who take part (Burkhalter et al., 2002; Warren, 1992). The participant effects attributed to robust deliberation include a heightened sense of political self-confidence and increased political activity beyond the particular issue under discussion (Pincock, 2012). Moreover, when a government agency or civic actor sponsoring a deliberation proves to be responsive to citizen input, participants may come to perceive that sponsor as more legitimate (Gastil et al., 2017) and may be motivated to become more politically active (Parent et al., 2005).

Linking Gamification and Deliberative Democratic Purposes

These four deliberative purposes we just reviewed relate in different ways to the four game mechanic categories identified by Lerner (2014). Table 1 organizes the extant literature on gamified civic engagement/consultation efforts to show how game mechanics can help achieve deliberative ends. It shows that each of the four game design categories has at least one plausible link to each of the four purposes of deliberation, with many more linkages omitted for simplicity. We do not limit our investigation to *online* applications of game design principles, but we will narrow our discussion in this way shortly. Also, some of these game elements have meaning outside of games. For instance, narratives play important roles not only in games but also in political discussions, protests, and other contexts (Polletta, 2006). Our point here is to think of these as elements of a well-designed game—not to miscategorize them as *only* game features.

Every example in Table 1 refers to a specific way that game design can change public behaviors or attitudes. It was harder to find examples of game elements shaping the behavior of public officials or the sponsors of public processes. The exception was that when agencies and community service providers were integrated into a deliberative process, this spurred a better understanding of public needs and responsiveness (Bista et al., 2014; Nepal et al., 2012; Rakhmawati and Fibrianto, 2016). In sum, there is ample evidence that game design elements connect with deliberative outcomes, at least for the principal participants in these games.

A Motivational Model of Online Civic Engagement

The studies summarized in Table 1 provide various accounts of the mechanisms by which gamified processes promote inclusion, deliberation, and other desired outcomes. From this point forward, however, we narrow our focus to look at *online* design tools and the motivational effects they can have on both participants in and government sponsors of such processes. In this section, we provide a theoretical overview of three fundamental

Table 1. Links Between Game Elements and Deliberative Process Features and Outcomes.

	Inclusive process with active participation	Quality of the deliberative process	High-quality input linked to policy outcomes	Long-term changes in civic attitudes and engagement
Conflict/ collaboration	*Leaderboards* encourage participation (Johannessen and Berntzen, 2016)	*Collaboration* promotes deliberation (Ampatzidou et al., 2018; Gordon and Schirra, 2011)	*Feedback loops* encourage better input (Münster et al., 2017)	*Competition* spurs ongoing civic activity (Rakhmawati and Fibrianto, 2016; Rehm et al., 2015)
Rules and norms	*Context-sensitive information* provision increases participation (Münster et al., 2017; Rehm et al., 2015)	*Social rating* promotes deliberation (Hu and Chen, 2015) and *fair play norms* encourage empathy (Gordon et al., 2017)	*Context-sensitive information* improves input (Münster et al., 2017)	*Context-sensitive information* about relevant upcoming events promotes civic engagement (Rehm et al., 2015)
Incremental progress	*Points/prizes/levels* sustain engagement (Fernandes and Junior, 2016; Kazhamiakin et al., 2016)	*Missions* generate more focused contributions (Thiel and Lehner, 2015) and collective discussion (Devisch et al., 2016)	*Points/levels* lead to higher-quality input (Rakhmawati and Fibrianto, 2016) and aid facilitation/ feedback (Bista et al., 2014)	*Points* can increase conventional political participation (Foxman and Forelle, 2014; Jhaver et al., 2017)
Engaging features	Compelling *narrative* (Prandi et al., 2017) and *social elements* sustain participation (Thiel, 2016; Thiel and Fröhlich, 2017)	*Concealing information* temporarily spurs deeper reflection (Gordon and Baldwin-Philippi, 2014)	*Playful environment* spurs contributions to local policy knowledge (Lindley and Coulton, 2015; Prandi et al., 2017)	*Visualization* elements enable participation in future urban planning (Münster et al., 2017)

behavioral motivations and then explain how each of these motivations connects to game design features.

Fundamental Motivations at Play

Motivation is a strong candidate for the mechanism by which game design tools shape deliberative processes. Public forums must motivate participants to both attend and engage (Dillard and Backhaus, 2007), and political systems need to spark public motivations to sustain deliberative fora (Conover et al., 2002). The antithesis of deliberative civic engagement may be "motivated reasoning," in which people reject information and arguments deemed incongruent with their preexisting biases (Jost and Amodio, 2011; Kahan, 2013; Leeper and Slothuus, 2014).

As for public agencies and officials, a frequent question among deliberative theorists and practitioners is the conditions motivating government to sponsor and respond to public engagement processes (Barrett et al., 2012; He and Warren, 2012; Knobloch et al., 2015;

Neblo et al., 2018). Though we focus principally on how game design can motivate public participants, later we will note the key motivational opportunities for government actors.

Theories of political behavior have motivational underpinnings that trace back to mid-century psychological theory—most prominently Abraham Maslow's "hierarchy of needs," which ascended from physical requirements through social and self-esteem needs to a kind of personal fulfillment called "self-actualization" (Fitzgerald, 1977). Modern evolutionary theories include the first three foundational categories plus other higher-order needs for mating and parenting (Kenrick et al., 2010; Schaller et al., 2017), but they subsume self-actualization under social status needs, self-esteem needs (Kenrick, 2017; Krems et al., 2017), and mate-seeking (Kenrick et al., 2010).

Though it is tempting to add to our list of motivations a need for unstructured play distinct from other physical, social, and esteem motivations, the resulting loss in theoretical parsimony might not yield a greater gain in explanatory power. After all, games are historically situated activities enacted in social contexts (Hakkarainen, 1999) linked to deeper affiliation motivations. Even solitaire games, including elaborate single-player online games, tap into players' achievement motivations as they progress through levels or exercise power in their games (Klimmt and Hartmann, 2006).

Motivational Civic Game Mechanics

Table 2 places the three motivational categories of physical, social, and esteem needs in a civic context, then connects them to the specific emotions facilitating them (Beall and Tracy, 2017) and proximate situational triggers (Kenrick et al., 2010). As an example, we divide affiliation motivations into desires to belong and empathy. We then trace both of these through situations, emotions, and game mechanics. In this case, the desire to belong would be felt more acutely by a person in an unfamiliar situation amid strangers. The person may feel lonely at first but then forge social connections to the others by using social gamification features, giving and receiving feedback, and embracing a shared narrative, even if it is fictional.

In the next section, we show how each game mechanic in Table 2 could improve the deliberative processes and outcomes of PB. We stress again that there exists a wider universe of game mechanics (Burak and Parker, 2017; Burke, 2014; Chou, 2015; Lerner, 2014; McGonigal, 2011; Salen and Zimmerman, 2004; Schell, 2014), but we highlight the subset of tools that have demonstrated their civic utility and appear to have sound motivational pedigrees.

Motivational Game Mechanics in PB

For a PB effort using digital tools, each of the specific game mechanics in Table 2 could be utilized through specific online tools to motivate PB participants. In the end of this section, we point out instances in which such motivational tools could induce public officials to enhance, expand, or sustain PB.

Meeting Basic Needs

PB allows participants to make decisions about resource allocation, often involving such basic infrastructural needs as adequate sanitation or such vital local services as public education (Wampler, 2007). This means that PB could appeal to the basic needs motivation—the first and most fundamental of the motivational categories (Schaller et al., 2017).

Table 2. Connecting Motivations, Situations, Emotions, and Game Mechanics.

	Subtype of motivation	Situational contributors	Initial and sought emotions	Game mechanics
Basic needs	Protecting material self-interest and loss aversion	External incentives linked to basic need fulfillment	Fear moving to security	Competition over resources Material rewards for effort
	Urgency of need	Time pressure for public decision	Anxiety/ desperation moving to relief	Scarcity of rewards Artificial time constraints
Affiliation	Desire to belong to a community	Familiar or unfamiliar places/ people Opportunities for social inclusion or exclusion	Loneliness and isolation moving to social cohesion	Social tools Peer rating system Shared narrative
	Empathy for the needs of others	Salience of social norms Cultural cues signaling shared identity and trust	Sympathy moving to happiness	Personal narratives Norms and rules Rewards for consensus and collective action
Status and esteem	Desire to learn or to "get it right"	Complexity and uncertainty in issues discussed	Sense of inadequacy moving to self-esteem	Learning modules Feasibility incentives
	Achievement drive for higher social standing	Lack of perceived respect Displays of dominance	Jealousy moving to pride	Progress bars, badges, and leveling Points and rewards Leaderboards Missions

The two game mechanics most directly associated with meeting basic needs are *competition over resources* and *scarcity of resources*, and they fit into two common components of online tools used in PB. First, different proposals in PB typically compete directly with each other, and the portion of the larger city budget allocated to PB is relatively small (Gilman, 2016). Second, competition over scarce resources in PB could manifest in reminding participants of the total costs of all proposals relative to the total budget available, encouraging larger proposals that could take greater shares of the total budget, and framing proposals as "winning" and "losing" in PB voting.

Typical PB processes, however, often emphasize cooperation and community bonding instead (Lerner, 2014). Such collaborative framing can promote deliberation (Ampatzidou et al., 2018; Gordon and Schirra, 2011), but the literature does not demonstrate a wide array of impacts in using this approach. In contrast, competitive game dynamics were the lone game mechanic in Table 1 directly associated with all four deliberative process features and outcome measures (e.g. Rehm et al., 2015; Tolmie et al., 2014).

Fortunately, two other game mechanics are associated with meeting basic needs too. Participants could receive *rewards for effort* during games, and this game mechanic could manifest in such game features as game systems translating in-game points into material

outcomes. Rewarding participants for their efforts during a game improved inclusion, deliberation, and policy outcomes (e.g. Bianchini et al., 2016b). In PB, however, this game mechanic could manifest in providing direct material rewards to participants for engaging in in-game activity, akin to payments made to jurors or participants in minipublics (e.g. Crosby and Nethercutt, 2005).

One's influence over PB allocations also could be linked on one's in-game activity: The greater one's in-game activity, the more weight one's vote would carry. Such a system of "sweat equity" could align well with democratic and activist impulses (Baiocchi et al., 2018), making this approach a more natural fit for PB in which participants' online participation in proposal development and evaluation is recorded automatically.

In addition, an online PB tool could create resource scarcity and competition that would pit participants not against each other but against "the clock." Such *artificial time constraints* are already built right into budgetary processes, which naturally involve meeting fiscal deadlines at every turn. Gamified deadlines could amplify a sense of time pressure throughout the entire process, while adding periodic artificial deadlines along the way. Though this game mechanic of racing against the clock is not common in the civic games literature, such ticking clocks or hourglasses are ubiquitous in games more generally (Schell, 2014: 165). Moreover, if missing a deadline would have negative consequences for all players, this game mechanic could become more cooperative and amenable to the conventional PB design.

Promoting Social Connection and Empathy

The second broad motivational category concerns meeting affiliative needs, which we split into two separate parts in Table 2. The first part concerns one's need to belong to one's community; three game mechanics relevant to this part are *social tools, peer ratings*, and *shared narratives*. In Table 1, we touched on the two ways of sustaining participation over time of utilizing social tools such as chat systems (e.g. Thiel and Fröhlich, 2017) and immersing participants in compelling shared narratives (Prandi et al., 2017). In much the same way, deliberation can benefit from both fictional narratives (Gordon and Manosevitch, 2010) and peer-rating systems (Hu and Chen, 2015).

These two ways of sustaining participation and improving deliberation could be combined in PB using online tools. Before delving into specific budget proposals, the online system could ask participants to reflect on the experiences of fictionalized residents of their community and what those individuals would need to achieve their life goals. Participants could discuss fictional scenarios based on real experiences of real residents and afterward give feedback on each other's discussion participation.

Promoting this motivation to meet affiliative needs would also involve inducing empathy in participants with residents as part of a shared community. *Sharing personal narratives*, whether real or fictional, can encourage perspective taking and mutual understanding, which are both helpful to deliberation (Gordon and Manosevitch, 2010). A more direct approach to encouraging deliberation would be introducing clear guidelines, such as *fair play norms* (Gordon et al., 2017) or *discussion rules* for deliberation (Gordon and Manosevitch, 2010).

Gamified environments can also utilize collaborative reward structures that provide additional incentives for reaching a consensus in deliberation or taking collective action (Hainey et al., 2016). Though we did not find such collaborative reward structures in the literature, online and mobile games lead players to collaborate on missions that require

carefully planned and coordinated collective action by multiple players, whether collaborating only temporarily or entering into more long-standing alliances (Madigan, 2015).

Such features could also complement other aforementioned affiliative features by establishing both clear guidelines and stronger incentives for online deliberation. In PB, participants' votes could carry greater weight if they could be part of a broader agreement or a supermajority for a consensus in deliberation among participants. Achieving such a supermajority for a consensus in deliberation among participants could also encourage the role of participants in PB as bodies of "trusted decision proxies" for residents (Warren and Gastil, 2015: 568). After randomly assigning community members to groups of 10, for example, each group could have its vote count double if at least 8 of its 10 members agreed on budget priorities.

Improving Self-Esteem and Social Status

The third motivational category combines one's needs to learn and improve one's social standing. This motivational category encompasses the greatest number of game mechanics in the far right column in Table 2, as multiple studies found that these associated game mechanics could improve all four deliberative process features and outcome measures (e.g. Bianchini et al., 2016a; Rakhmawati and Fibrianto, 2016).

The most straightforward of these game mechanics—*learning modules* and *context-sensitive information*—utilize what Druckman (2012) calls the "accuracy motivation." People often engage in biased reasoning processes that can even lead to "boomerang" effects, whereby corrective information reinforces existing biases (Hart and Nisbet, 2012). Nonetheless, repeated exposure to disconfirming information can create sufficient anxiety even in committed partisans that they adjust their beliefs in response (Redlawsk et al., 2010). More generally, when reliable information about candidates and ballot measures is provided in a straightforward manner, many voters can absorb such data into their beliefs before casting their ballots (Lau and Redlawsk, 2006; Már and Gastil, 2019). Game mechanics in Table 1 associated with this motivational category involve *concealing information temporarily* to refocus deliberation (Gordon and Baldwin-Philippi, 2014) and providing tools for *visualization* to make the information more accessible (Münster et al., 2017).

More powerful still could be pairing this accuracy motivation with the motivation for achievement. Players often desire to "get it right" in games because of the rewards at stake. If financial incentives are strong enough, simple prediction games can motivate ideological groups to more accurately forecast future events (e.g. the fiscal impact of new tax laws) even when their predictions contradict their own ideologies (Tetlock et al., 2014). Table 1 establishes that online civic engagement can effectively utilize lesser rewards, such as allowing players to see their *progress bars* move forward or inviting them to *level up* in games to unlock new abilities or opportunities. Online games also use more direct rewards or grant in-game points for completing in-game activities (e.g. Fernandes and Junior, 2016; Münster et al., 2017). Badges and leaderboards can broadcast one's point accumulation in a way that satisfies one's need for social recognition (Devisch et al., 2016), and in-game results being made visible to others outside the game can serve as a recruitment tool for other players for the game (Johannessen and Berntzen, 2016).

In a PB process, accuracy and social status motivations could impel participants to learn more information about budget proposals, the larger budget, and the implementation phase of PB. By offering rewards for completing learning modules, PB efforts could encourage

participants to be more sophisticated producers and evaluators of budget proposals and follow the long-term processes of implementing the successful ones. For instance, participants might covet "feasibility points" granted in two stages. First, when draft proposals receive favorable feasibility reviews from local engineers, the proposals' drafters could receive feasibility points. Second, when funded proposals are completed successfully on or under budget, both those who drafted it and voted for it could receive more feasibility points.

Finally, the *missions* game mechanic incorporates accuracy and social status motivations by providing players clear challenges they must complete to receive rewards (e.g. Thiel and Lehner, 2015). In a PB process, for example, participants could receive "good neighbor" missions requiring them to "check in" at public meetings, workshops, and civic events during crucial phases in the PB process. Only those participants who successfully complete the mission would gain the rewards, and those participants who attend those events would also gain knowledge and social connections they would have otherwise missed.

Motivating Public Officials

The only game mechanics listed in Table 1 not yet mentioned so far are *feedback loops* (e.g. Crowley et al., 2012). We saved those for this section because the role of feedback is important in theorizing what would motivate public officials in the context of online public consultation or civic engagement. As noted earlier, two prior studies found that when government agencies or community service providers were successfully integrated into public consultation through continuously monitoring and reciprocating public input, these entities could understand and respond to public needs more effectively (Bista et al., 2014; Rakhmawati and Fibrianto, 2016). Feedback loops are also central to theories of digital civic engagement (e.g. Gastil and Richards, 2017) and public deliberation more generally (Burkhalter et al., 2002).

The power of these feedback loops comes from combining them with those other previously mentioned game mechanics and corresponding motivations. For example, consider the proposed "feasibility points" game mechanic mentioned before. Rewarding participants for producing proposals that proved feasible (on government review or through their implementation) creates a feedback loop with the feedback coming either directly from government or indirectly from the larger process of project implementation.

This process could include motivational features for the government agencies conducting initial feasibility ratings of project proposals. Any agency that makes their assessments of project proposals during the implementation process could receive rewards. At a minimum, the office could get "good estimate" badges, which might boost public confidence not only in the agency's future assessments but perhaps in the agency itself. If local government was paying third-party vendors for the software used to take PB processes online, perhaps accurate ratings could earn agencies discounts on future uses of that software.

A more ambitious implementation of a feedback loop could allow PB's influence and legitimacy to expand gradually. In this scenario, city government could begin with modest demonstration budgets for taking PB online, as has happened in cities like New York (Gilman, 2016). At the end of the PB process, participants and government would both evaluate the overall quality of both deliberation leading up to the vote and choices of the final project proposal. If both sides provided sufficiently favorable supermajority marks—like a two-thirds supermajority approval among both city staff and PB

participants, for example—that would likely induce an expansion of PB's influence and legitimacy. The net result of this feedback loop in PB could be a boost not only in the political self-confidence of city residents but also in legitimacy accorded to city government from those same residents.

Refining a Theoretical Model Through Multi-Stage Experimentation

We have presented a series of empirical propositions on the civic benefits of adding specific online game elements to a PB system. In this final section, we argue for testing those claims with a three-step experimental approach. Experiments have proven fruitful in prior studies of public deliberation (Gastil, 2018), though truly randomized and controlled trials remain rare (for an exception, see Carman et al., 2015). In much the same way, prior literature reviews of game mechanics have lamented the lack of experimental trials of game mechanics' effectiveness as educational tools (Hainey et al., 2016).

Thus, we recommend an experimental approach beginning with demonstrations of external and ecological validity in applied settings and moving after toward more fine-grained experimentation clarifying individual effects and motivational dynamics of more specific game elements. The first step would gauge the overall soundness of our experimental approach by testing an established PB system using online tools against an alternative that features several key game mechanics. This gamified variant could include, for instance, a larger narrative describing a need for residents to revitalize their neighborhood. To reach that goal, participants would need to collaborate in teams to develop project proposals that would compete with each other for limited funding. At different steps along the way, participants could interact with their neighbors and city officials to earn in-game points, level up their in-game personas, earn badges for completing missions, and rise up the leaderboards.

By design, the proposed system would include multiple game elements woven together into one coherent and whole system, with the point of the experiment being to see whether such a complete game would yield different outcomes. Such a test would include periodic surveys to ascertain participants' motivations, emotional states, and perceptions of the game itself as it progresses from idea generation to voting on proposals. Ideally, such a game would continue through two iterations of PB. This way, the system's feedback loops connecting residents and city officials would be used enough times to yield significant increases in broader civic attitudes, such as trust in one's city government or one's neighbors.

If such a larger test proved fruitful, the second phase of experimentation would more closely analyze individual game mechanics. Data from the first step including qualitative feedback from both participants and public officials would prioritize which game mechanics to analyze more closely. The point of these experiments, which could occur both in the field and in lab environments, would be to refine the most promising game mechanics and explore their interdependence. In this essay, we presented each game mechanic as though it operated independently of the rest, but the benefits of one mechanic may ultimately depend on the presence of another. For instance, it would be fruitless to add a collaborative game mechanic into a system that lacked the social tools necessary for planning and coordination.

After making refinements to the online tools in a PB system in light of these experiments' findings, the third and final step would return to the field for randomized controlled trials. In time, these tests could be conducted across a range of different settings to see whether the online system functions well across diverse sociopolitical settings or

beyond the PB process itself. For instance, could such an online system sustain itself over time and continue producing desired deliberative effects—even when facing external challenges, such as organized opposition to its operation or its decisions? Also, since only a subgroup of the population has begun to live "permanently online" (Vorderer et al., 2017), will a PB process using online tools exacerbate existing demographic divides by privileging those already more active online?

From our theoretical standpoint, the ultimate question for a PB system is whether it will enhance the quality of life for the communities it serves. Thus, we need to know whether incorporating online tools into a PB system will improve residents' living conditions and create more opportunities for them to thrive. Will it enable communities to govern themselves more effectively? Such queries lie far down the road, but we hope to have provided guidance on the steps one could take toward refining such questions and answering them through rigorous field tests of a PB system incorporating gamified online tools.

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ORCID iD

John Gastil https://orcid.org/0000-0002-0337-1064

References

Ampatzidou C, Gugerell K, Constantinescu T, et al. (2018) All Work and No Play? Facilitating Serious Games and Gamified Applications in Participatory Urban Planning and Governance. *Urban Planning* 3 (1): 34–46.

Baek YM, Wojcieszak M and Delli Carpini MX (2012) Online Versus Face-To Deliberation: Who? Why? What? With What Effects? *New Media & Society* 14 (3): 363–383.

Baiocchi G and Ganuza E (2014) Participatory Budgeting as if Emancipation Mattered. *Politics & Society* 42 (1): 29–50.

Baiocchi G, Brady M and Carlson HJ (2018) Beyond the Market: Housing Alternatives from the Grassroots. *Dissent* 65 (4): 51–58.

Barrett G, Wyman M and Coelho VSP (2012) Assessing the Policy Impacts of Deliberative Civic Engagement in the Health Policy Processes of Brazil and Canada. In: Nabatchi T, Gastil J, Weiksner M, et al. (eds) *Democracy in Motion: Evaluating the Practice and Impact of Deliberative Civic Engagement*. New York: Oxford University Press, pp.181–203.

Barros SAR and Sampaio RC (2016) Do Citizens Trust Electronic Participatory Budgeting? Public Expression in Online Forums as an Evaluation Method in Belo Horizonte. *Policy & Internet* 8 (3): 292–312.

Beall AT and Tracy JL (2017) Emotivational Psychology: How Distinct Emotions Facilitate Fundamental Motives. *Social & Personality Psychology Compass* 11 (2): e12303.

Berg J (2016) The Impact of Anonymity and Issue Controversiality on the Quality of Online Discussion. *Journal of Information Technology & Politics* 13 (1): 37–51.

Bianchini D, Fogli D and Ragazzi D (2016a) Promoting Citizen Participation through Gamification. In: *Proceedings of the 9th Nordic Conference on Human—Computer Interaction—NordiCHI '16*, *Gothenburg*, 23–27 October, pp.1–4. New York: ACM.

Bianchini D, Fogli D and Ragazzi D (2016b) TAB Sharing: A Gamified Tool for E-Participation. In: *Proceedings of the International Working Conference on Advanced Visual Interfaces—AVI '16*, *Bari*, 7–10 June, pp.294–295. New York: ACM.

Bimber B (1998) The Internet and Political Transformation: Populism, Community, and Accelerated Pluralism. *Polity* 31: 133–160.

Bista SK, Nepal S, Paris C, et al. (2014) Gamification for Online Communities: A Case Study for Delivering Government Services. *International Journal of Cooperative Information Systems* 23: 1441002.

Boyle EA, Hainey T, Connolly TM, et al. (2016) An Update to the Systematic Literature Review of Empirical Evidence of the Impacts and Outcomes of Computer Games and Serious Games. *Computers & Education* 94: 178–192.

Burak A and Parker L (2017) *Power Play: How Video Games Can Save the World*. New York: St. Martin's Press.

Burke B (2014) *Gamify: How Gamification Motivates People to Do Extraordinary Things*. Brookline, MA: Bibliomotion.

Burkhalter S, Gastil J and Kelshaw T (2002) A Conceptual Definition and Theoretical Model of Public Deliberation in Small Face-To Groups. *Communication Theory* 12 (4): 398–422.

Carman KL, Mallery C, Maurer M, et al. (2015) Effectiveness of Public Deliberation Methods for Gathering Input on Issues in Healthcare: Results from a Randomized Trial. *Social Science & Medicine* 33: 11–20.

Chou Y (2015) *Actionable Gamification: Beyond Points, Badges, and Leaderboards*. Fremont, CA: Octalysis Media.

Cohen J (1997) Deliberation and Democratic Legitimacy. In: Bohman JF and Rehg W (eds) *Deliberative Democracy: Essays on Reason and Politics*. Cambridge, MA: MIT Press, pp.67–91.

Conover PJ, Searing DD and Crewe IM (2002) The Deliberative Potential of Political Discussion. *British Journal of Political Science* 32: 21–62.

Crosby N and Nethercutt D (2005) Citizens Juries: Creating a Trustworthy Voice of the People. In: Gastil J and Levine P (eds) *The Deliberative Democracy Handbook: Strategies for Effective Civic Engagement in the Twenty-first Century*. San Francisco, CA: Jossey-Bass, pp.111–119.

Crowley DN, Breslin JG, Corcoran P, et al. (2012) Gamification of Citizen Sensing through Mobile Social Reporting. In: *Proceedings of the 2012 IEEE International Games Innovation Conference*, *Rochester, NY*, 7–9 September 2012, pp.1–5. New York: IEEE.

Dahl RA (1989) *Democracy and Its Critics*. New Haven, CT: Yale University Press.

Davies T and Chandler R (2012) Online Deliberation Design: Choices, Criteria, and Evidence. In: Nabatchi T, Gastil J, Weiksner M, et al. (eds) *Democracy in Motion: Evaluating the Practice and Impact of Deliberative Civic Engagement*. New York: Oxford University Press, pp.103–133.

Devisch O, Poplin A and Sofronie S (2016) The Gamification of Civic Participation: Two Experiments in Improving the Skills of Citizens to Reflect Collectively on Spatial Issues. *Journal of Urban Technology* 23 (2): 81–102.

Dias N (ed) (2018) *Hope for Democracy: 30 Years of Participatory Budgeting Worldwide*. New York: Participatory Budgeting Project.

Dillard JP and Backhaus SJ (2007) Civic Deliberation, Emotional Response, and Political Involvement. *The International Journal of Public Participation* 1 (2): 1–18.

Druckman JN (2012) The Politics of Motivation. *Critical Review* 24 (2): 199–216.

Ercan SA, Hendriks CM and Dryzek JS (2018) Public Deliberation in an Era of Communicative Plenty. *Politics and Policy* 47 (1): 19–36.

Fernandes FT and Junior PTA (2016) Gamification Aspects in the Context of Electronic Government and Education: A Case Study. In: Nah FF-H and Tan C-H (eds) *HCI in Business, Government, and Organizations: Information Systems*. New York: Springer, pp.140–150.

Fitzgerald R (1977) *Human Needs and Politics*. Amsterdam: Elsevier.

Foxman M and Forelle M (2014) Electing to Play: MTV's Fantasy Election and Changes in Political Engagement through Gameplay. *Games and Culture* 9 (6): 454–467.

Ganuza E and Francés F (2012) The Deliberative Turn in Participation: the Problem of Inclusion and Deliberative Opportunities in Participatory Budgeting. *European Political Science Review* 4 (2): 283–302.

Gastil J (2000) Is Face-To Citizen Deliberation a Luxury or a Necessity? *Political Communication* 17 (4): 357–361.

Gastil J (2018) The Lessons and Limitations of Experiments in Democratic Deliberation. *Annual Review of Law and Social Science* 14: 271–291.

Gastil J and Richards RC (2017) Embracing Digital Democracy: A Call for Building an Online Civic Commons. *PS: Political Science & Politics* 50 (3): 758–763.

Gastil J, Knobloch K and Kelly M (2012) Evaluating Deliberative Public Events and Projects. In: Nabatchi T, Gastil J, Weiksner M, et al. (eds). *Democracy in Motion: Evaluating the Practice and Impact of Deliberative Civic Engagement*. New York: Oxford University Press, pp.205–229.

Gastil J, Richards R, Ryan M, et al. (2017) Testing Assumptions in Deliberative Democratic Design: A Preliminary Assessment of the Efficacy of the Participedia Data Archive as an Analytic Tool. *Journal of Public Deliberation* 13: 2.

Giessen HW (2015) Serious Games Effects: An Overview. *Procedia—Social and Behavioral Sciences* 174: 2240–2244.

Gilman HR (2016) *Democracy Reinvented: Participatory Budgeting and Civic Innovation in America*. Washington, DC: Brookings Institution Press.

Gordon E and Baldwin-Philippi J (2014) Playful Civic Learning: Enabling Reflection and Lateral Trust in Game-Based Public Participation. *International Journal of Communication* 8: 759–786.

Gordon E and Manosevitch E (2010) Augmented Deliberation: Merging Physical and Virtual Interaction to Engage Communities in Urban Planning. *New Media & Society* 13: 1–21.

Gordon E and Schirra S (2011) Playing with Empathy: Digital Role-Playing Games in Public Meetings. In: *Proceedings of the 5th International Conference on Communities and Technologies—C&T '11*, *Brisbane, QLD, Australia*, 29 June–2 July 2011, p.179. New York: ACM.

Gordon E, Haas J and Michelson B (2017) Civic Creativity: Role-Playing Games in Deliberative Process. *International Journal of Communication* 11: 19.

Gordon E and Mihailidis P (eds) (2016) *Civic Media: Technology, Design, Practice*. Cambridge, MA: MIT Press.

Green DP and Gerber AS (2008) *Get Out the Vote: How to Increase Voter Turnout*. Washington, DC: Brookings Institution Press.

Hainey T, Connolly TM, Boyle EA, et al. (2016) A Systematic Literature Review of Games-Based Learning Empirical Evidence in Primary Education. *Computers & Education* 102: 202–223.

Hakkarainen P (1999) Play and Motivation. In: Engeström Y, Miettinen R, and Punamäki R-L (eds) *Perspectives on Activity Theory*. New York: Cambridge University Press, pp.231–249.

Haller C and Faulkner G (2012) Participatory Budgeting in Denver, Colorado. *National Civic Review* 3 (101): 23–25.

Hart PS and Nisbet EC (2012) Boomerang Effects in Science Communication: How Motivated Reasoning and Identity Cues Amplify Opinion Polarization about Climate Mitigation Policies. *Communication Research* 39 (6): 701–723.

Hassan L (2017) Governments Should Play Games: Towards a Framework for the Gamification of Civic Engagement Platforms. *Simulation & Gaming* 48 (2): 249–267.

Hassan L and Hamari J (2019) Gamification of E-Participation: A Literature Review. In: *Proceedings of the 52nd Hawaii International Conference on System Sciences*, 5–8 January, pp.3077–3086. Available at: https://scholarspace.manoa.hawaii.edu/bitstream/10125/59744/0304.pdf (accessed 20 November 2019).

He B and Warren ME (2012) Authoritarian Deliberation: The Deliberative Turn in Chinese Political Development. *Perspectives on Politics* 9 (2): 269–289.

Howard PN (2015) *Pax Technica: How the Internet of Things May Set Us Free or Lock Us Up*. New Haven, CT: Yale University Press.

Hu S-C and Chen I-C (2015) A Gamified Online Forum Inspiring Group Intelligence Distillation for Policy Making. In: Tan Y, Shi Y, Buarque F, et al. (eds) *Advances in Swarm and Computational Intelligence*. New York: Springer, pp.423–430.

Jhaver S, Vora P and Bruckman A (2017) *Designing for Civil Conversations: Lessons Learned from Changemyview*. Atlanta, GA: Georgia Institute of Technology.

Johannessen MR and Berntzen L (2016) Smart Cities through Implicit Participation: Using Gamification to Generate Citizen Input for Public Transport Planning. *Innovation and the Public Sector*:: 23–30.

Johnson AM and Roman A (2015) Reflections on E-Rulemaking: Challenges, Limitations and Unrealistic Expectations. *The Electronic Journal of E-government* 13 (1): 43–55.

Johnson C and Gastil J (2015) Variations of Institutional Design for Empowered Deliberation. *Journal of Public Deliberation* 11: 2.

Johnson GF (2009) Deliberative Democratic Practices in Canada: An Analysis of Institutional Empowerment in Three Cases. *Canadian Journal of Political Science* 42: 679–703.

Johnson GF (2015) *Democratic Illusion: Deliberative Democracy in Canadian Public Policy*. Toronto: University of Toronto Press.

Jost JT and Amodio DM (2011) Political Ideology as Motivated Social Cognition: Behavioral and Neuroscientific Evidence. *Motivation and Emotion* 36 (1): 55–64.

Kahan DM (2013) Ideology, Motivated Reasoning, and Cognitive Reflection. *Judgment and Decision Making* 8 (4): 407–424.

Kazhamiakin R, Marconi A, Martinelli A, et al. (2016) A Gamification Framework for the Long-Term Engagement of Smart Citizens. In: *Proceedings of the 2016 IEEE International Smart Cities Conference ISC220162 (ISC2)*, *Trento*, 12–15 September, pp.1–7. New York: IEEE.

Kenrick DT (2017) Self-Actualization, Human Nature, and Global Social Problems. *Society* 54 (6): 520–523.

Kenrick DT, Griskevicius V, Neuberg SL, et al. (2010) Renovating the Pyramid of Needs: Contemporary Extensions Built upon Ancient Foundations. *Perspectives on Psychological Science* 5 (3): 292–314.

Klimmt C and Hartmann T (2006) Effectance, Self-Efficacy, and the Motivation to Play Video Games. In: Vorderer P and Bryant J (eds) *Playing Video Games: Motives, Responses, and Consequences*. Abingdon: Routledge, pp.133–145.

Knobloch KR, Gastil J and Reitman T (2015) Connecting Micro-deliberation to Electoral Decision Making: Institutionalizing the Oregon CIR. In: Przybylska A (ed) *Deliberation and Democracy: Innovative Processes and Institutions* (Warsaw Studies in Politics and Society). New York: Peter Lang, pp.21–40.

Krems JA, Kenrick DT and Neel R (2017) Individual Perceptions of Self-Actualization: What Functional Motives Are Linked to Fulfilling One's Full Potential? *Personality and Social Psychology Bulletin* 43 (9): 1337–1352.

Lau RR and Redlawsk DP (2006) *How Voters Decide: Information Processing during Election Campaigns*. New York: Cambridge University Press.

Leeper TJ and Slothuus R (2014) Political Parties, Motivated Reasoning, and Public Opinion Formation. *Political Psychology* 35: 129–156.

Lerner J (2014) *Making Democracy Fun: How Game Design Can Empower Citizens and Transform Politics*. Cambridge, MA: MIT Press.

Lim S and Oh Y (2016) Online Versus Offline Participation: Has the Democratic Potential of the Internet Been Realized? Analysis of a Participatory Budgeting System in Korea. *Public Performance & Management Review* 39 (3): 676–700.

Lindley J and Coulton P (2015) Game of Drones. In: *Proceedings of the 20 Annual Symposium on Computer—Human Interaction in Play—CHI PLAY '15*, *London*, 5–7 October, pp.613–618. New York: ACM.

Lyons B (2017) From Code to Discourse: Social Media and Linkage Mechanisms in Deliberative Systems. *Journal of Public Deliberation* 13: 4.

McGonigal J (2011) *Reality Is Broken: Why Games Make Us Better and How They Can Change the World*. New York: Penguin Books.

Madigan J (2015) *Getting Gamers: The Psychology of Video Games and Their Impact on the People Who Play Them*. Lanham, MD: Rowman & Littlefield.

Manosevitch I (2014) The Design of Online Deliberation: Implications for Practice, Theory and Democratic Citizenship. *Journal of Public Deliberation* 10 (1): 9.

Már K and Gastil J (2019) Tracing the Boundaries of Motivated Reasoning: How Deliberative Minipublics Can Improve Voter Knowledge. *Political Psychology*. Available at: https://onlinelibrary.wiley.com/doi/abs/10.1111/pops.12591

Mayer IS (2009) The Gaming of Policy and the Politics of Gaming: A Review. *Simulation & Gaming* 40 (6): 825–862.

Meloni M, Allegretti G and Antunes S (2018) Participlaying: a Reflection on Gamification Techniques from the Standpoint of Participatory Budgeting. In: Dias N (ed) *Hope for Democracy: 30 Years of Participatory Budgeting Worldwide*. São Brás de Alportel: In Loco Association, pp.566–587.

Min S-J (2007) Online Vs. Face-To Deliberation: Effects on Civic Engagement. *Journal of Computer-mediated Communication* 12 (4): 1369–1387.

Miori V and Russo D (2011) Integrating Online and Traditional Involvement in Participatory Budgeting. *Electronic Journal of E-government* 9 (1): 41–57.

Morschheuser B, Hamari J, Koivisto J, et al. (2017) Gamified Crowdsourcing: Conceptualization, Literature Review, and Future Agenda. *International Journal of Human-computer Studies* 106: 26–43.

Münster S, Georgi C, Heijne K, et al. (2017) How to Involve Inhabitants in Urban Design Planning by Using Digital Tools? An Overview on a State of the Art, Key Challenges and Promising Approaches. *Procedia Computer Science* 112: 2391–2405.

Myers CD, Gordon HG, Kim HM, et al. (2018) Does Group Deliberation Mobilize? The Effect of Public Deliberation on Willingness to Participate in Politics. *Political Behavior* 1: 1–24.

Neblo MA, Esterling KM and Lazer DM (2018) *Politics with the People: Building a Directly Representative Democracy*. New York: Cambridge University Press.

Nepal S, Bista S, Colineau N, et al. (2012) Using Gamification in an Online Community. In: *Proceedings of the 8th IEEE International Conference on Collaborative Computing: Networking, Applications and Worksharing*, *Pittsburgh, PA*, 14–17 October.

Niemeyer S and Dryzek JS (2007) The Ends of Deliberation: Meta-consensus and Inter-Subjective Rationality as Ideal Outcomes. *Swiss Political Science Review* 13 (4): 497–526.
Nilsson B and Carlsson E (2014) Swedish Politicians and New Media: Democracy, Identity and Populism in a Digital Discourse. *New Media & Society* 16 (4): 655–671.
Noveck BS (2015) *Smart Citizens, Smarter State: The Technologies of Expertise and the Future of Governing*. Cambridge, MA: Harvard University Press.
Parent M, Vandebeek CA and Gemino AC (2005) Building Citizen Trust through E-Government. *Government Information Quarterly* 22 (4): 720–736.
Parkinson J and Mansbridge J (eds) (2012) *Deliberative Systems: Deliberative Democracy at the Large Scale*. New York: Cambridge University Press.
Peixoto T (2009) Beyond Theory: E-Participatory Budgeting and Its Promises for EParticipation. *European Journal of Epractice* 7 (5): 1–9.
Persily N (2017) The 2016 U.S. Election: Can Democracy Survive the Internet? *Journal of Democracy* 28 (2): 63–76.
Pincock H (2012) Does Deliberation Make Better Citizens? In: Nabatchi T, Gastil J, Weiksner M, et al. (eds) *Democracy in Motion: Evaluating the Practice and Impact of Deliberative Civic Engagement*. New York: Oxford University Press, pp.135–162.
Piotrowski SJ (2017) The "open Government Reform" Movement: The Case of the Open Government Partnership and U.S. Transparency Policies. *American Review of Public Administration* 47 (2): 155–171.
Polletta F (2006) *It Was Like a Fever: Storytelling in Protest and Politics*. Chicago: University of Chicago Press.
Porumbescu GA (2017) Does Transparency Improve Citizens' Perceptions of Government Performance? Evidence from Seoul, South Korea. *Administration & Society* 49 (3): 443–468.
Prandi C, Roccetti M, Salomoni P, et al. (2017) Fighting Exclusion: A Multimedia Mobile App with Zombies and Maps as a Medium for Civic Engagement and Design. *Multimedia Tools and Applications* 76 (4): 4951–4979.
Rakhmawati NA and Fibrianto B (2016) Designing a Gamification for Monitoring Surabaya City Development. In: *Proceedings of the 2016 International Conference on Information Communication Technology and Systems (ICTS)*, *Surabaya, Indonesia*, 12 October 2016, pp.262–265. New York: IEEE.
Redlawsk DP, Civettini AJW and Emmerson KM (2010) The Affective Tipping Point: Do Motivated Reasoners Ever 'get It'? *Political Psychology* 31 (4): 563–593.
Rehm S, Foth M and Mitchell P (2015) Gamifying Collective Intelligence for the Common Good. In: *Proceedings of the Workshop "Encouraging Collective Intelligence for the Common Good"* (eds Schuler D, De Cindio F and De Liddo A), *Limerick*, 27–30 June 2015.
Salen K and Zimmerman E (2004) *Rules of Play: Game Design Fundamentals*. Cambridge, MA: MIT Press.
Schaller M, Kenrick DT, Neel R, et al. (2017) Evolution and Human Motivation: A Fundamental Motives Framework. *Social and Personality Psychology Compass* 11 (6): e12319.
Schattan P, Coelho V, Pozzoni B, et al. (2005) Participation and Public Policies in Brazil. In: Gastil J and Levine P (eds) *The Deliberative Democracy Handbook: Strategies for Effective Civic Engagement in the Twenty-First Century*. San Francisco, CA: Jossey-Bass, pp.174–184.
Schell J (2014) *The Art of Game Design: A Book of Lenses*. 2nd ed. Boca Raton, FL: CRC Press.
Simone MA (2010) Deliberative Democracy Online: Bridging Networks with Digital Technologies. *The Communication Review* 13: 120–139.
Smith A (2018) Digital Platforms for Urban Democracy? In: *Proceedings of the conference Toward a Platform Urbanism Agenda for Urban Studies*, *University of Sheffield, Sheffield*, 3 September.
Souza C (2007) Local Democratization in Brazil: Strengths and Dilemmas of Deliberative Democracy. *Development* 50: 90–95.
Spada P (2012) Brazilian Participatory Budgeting Census: 1989-2012. Available at: https://dataverse.harvard.edu/dataverse/Participedia (accessed 20 April 2017).
Spada P, Allegretti G, Secchi M, et al. (2016) *Integrating Multiple Channels of Engagement: From Multichannel Marketing to Democratic Innovations*. Unpublished report. Available at: academia.edu.
Stromer-Galley J (2002) New Voices in the Public Sphere: A Comparative Analysis of Interpersonal and Online Political Talk. *Javnost the Public* 9: 23–42.
Su C (2017) From Porto Alegre to New York City: Participatory Budgeting and Democracy. *New Political Science* 39 (1): 67–75.
Sunstein CR (2017) *#Republic: Divided Democracy in the Age of Social Media*. Princeton, NJ: Princeton University Press.

Tetlock PE, Mellers BA, Rohrbaugh N, et al. (2014) Forecasting Tournaments: Tools for Increasing Transparency and Improving the Quality of Debate. *Current Directions in Psychological Science* 23 (4): 290–295.

Thiel S-K (2016) Reward-Based Vs. Social Gamification: Exploring Effectiveness of Gamefulness in Public Participation. In: *Proceedings of the 9th Nordic Conference on Human—Computer Interaction—NordiCHI '16*, *Gothenburg*, 23–27 October 2016, pp.1–6. *New York: ACM*.

Thiel S-K and Fröhlich P (2017) Gamification as Motivation to Engage in Location-Based Public Participation? In: Gartner G and Huang H (eds) *Progress in Location-Based Services 2016*. Cham: Springer, pp.399–421.

Thiel S-K and Lehner U (2015) Exploring the Effects of Game Elements in m-Participation. In: *Proceedings of the 20 British HCI Conference—British HCI '15*, *Lincoln*, 13–17 July, pp.65–73. New York: ACM.

Thiel S-K, Reisinger M, Röderer K, et al. (2016) Playing (with) Democracy: A Review of Gamified Participation Approaches. *eJournal of eDemocracy and Open Government: JeDEM* 8 (3): 32–60.

Tolmie P, Chamberlain A and Benford S (2014) Designing for Reportability: Sustainable Gamification, Public Engagement, and Promoting Environmental Debate. *Personal and Ubiquitous Computing* 18 (7): 1763–1774.

Vorderer P, Hefner D, Reinecke L, et al. (2017) *Permanently Online, Permanently Connected: Living and Communicating in a POPC World*. Abingdon: Routledge.

Wampler B (2007) *Participatory Budgeting in Brazil: Contestation, Cooperation, and Accountability*. University Park, PA: Pennsylvania State University Press.

Warren ME (1992) Democratic Theory and Self-Transformation. *American Political Science Review* 86 (1): 8–23.

Warren ME and Gastil J (2015) Can Deliberative Minipublics Address the Cognitive Challenges of Democratic Citizenship? *Journal of Politics* 77 (2): 562–574.

Wells C (2015) *The Civic Organization and the Digital Citizen: Communicating Engagement in a Networked Age*. New York: Oxford University Press.

Author Biographies

John Gastil (PhD, University of Wisconsin–Madison) is a professor in the Department of Communication Arts and Sciences and Political Science at the Pennsylvania State University, where he is a senior scholar at the McCourtney Institute for Democracy. His most recent books include *Legislature by Lot* and a second edition of *Democracy in Small Groups.*

Michael Broghammer (BA, Santa Clara University) is a graduate student in the Department of Communication Arts and Sciences and Political Science at the Pennsylvania State University. He studies how democratic deliberation and civic engagement may be inspired and facilitated in innovative and efficacious ways by new modern media and online digital technologies.